Cluster Analysis

Retail transaction data is widely used in clustering analysis because it helps uncover meaningful patterns in customer behavior and product demand.

In this dataset, clustering is applied in two key areas:

1. Customer Segmentation
Clustering is used to group customers based on their purchasing behavior, such as how often they buy, how much they spend, and how recently they made a purchase. This helps businesses identify different types of customers, such as loyal customers, occasional buyers, or those at risk of churning, enabling more targeted marketing strategies.

2. Demand Pattern Analysis
Clustering is also used to analyze product demand patterns. By grouping items based on sales behavior—such as frequency, quantity, and variability—businesses can identify fast-moving products, seasonal items, and slow-moving inventory. This insight supports better inventory management and demand forecasting.

1. Customer Segmentation (RFM Clustering)

Step by step clustering workflow:

- ☑ Data Cleaning. Remove all the missing values and duplicates and handle incorrect values (Example: Negative quantities , Invalid prices)
- ☑ Transform the cleaned dataset by grouping the data by 'Customer ID' and computing the RFM (Recency, Frequency, Monetary) metrics.
- ☑ Feature Scaling. Clustering algorithms depend on distance calculations. Therefore scaling is essential. Here we used Standardization method (mean = 0, standard deviation = 1).
- ☑ Choose a clustering algorithm. Used the most common clustering technique **'K-Means'** for customer segmentation.

- ☑ Determine the optimal number of clusters. Used the Elbow method.
- ☑ Train the clustering method using K-Means. Now every customer belongs to a cluster.
- ☑ Cluster Visualization. Used the PCA technique to reduce the dementianality to two.
- ☑ Cluster Profiling. It is the most important step to understand what each cluster represents.
- ☑ Cluster Interpretation.

RFM Features:

Feature	Meaning	Formulas
Recency (R)	How recently a customer purchased	Reference Date – Last Purchase Date
Frequency (F)	How many purchases the customer made	Number of transactions
Monetary (M)	How much money the customer spent	Total money spent

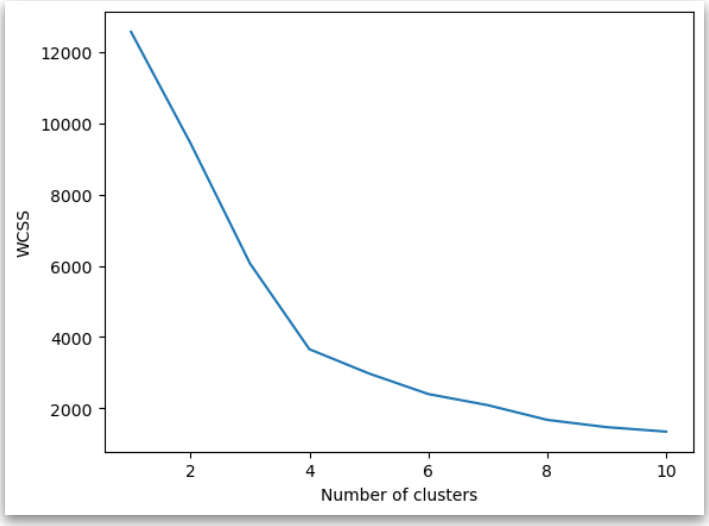
Recency (R) → lower = more recent (better)

Frequency (F) → higher = buys more often

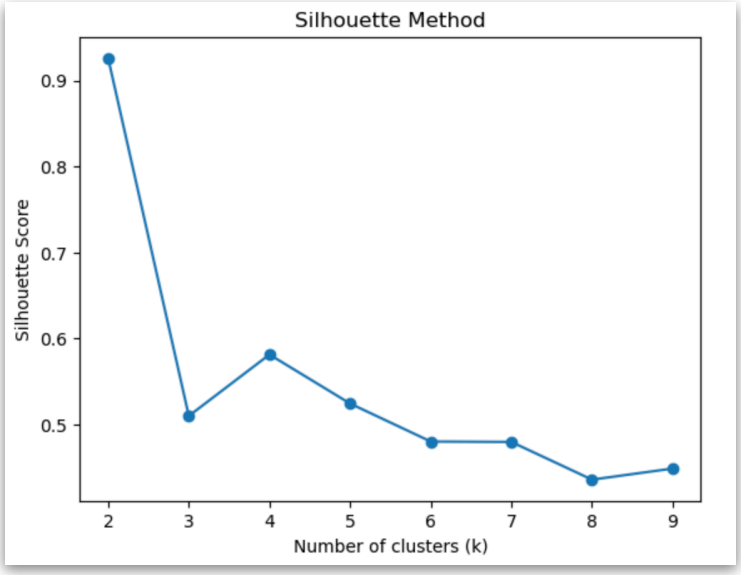
Monetary (M) → higher = spends more

Determine the optimal number of clusters.

Elbow Method



Silhouette Score



Mathematically, choosing 2 clusters yields the most compact and well-separated groups. However, for business analysis, selecting a higher number of clusters can provide more meaningful insights. Therefore, we chose 4 clusters to ensure the segments are actionable and better reflect the diversity of customer behavior.

Cluster Visualization

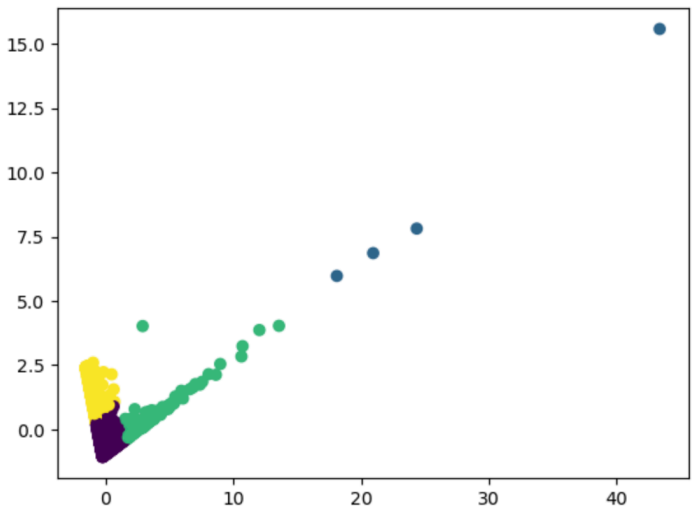
```
# Cluster Visualization
# Using PCA (dimensionality reduction)

from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

pca = PCA(n_components=2)
rfm_pca = pca.fit_transform(rfm_scaled)
print(rfm_pca.shape)

plt.scatter(rfm_pca[:,0], rfm_pca[:,1], c=rfm["Cluster"])
plt.show()
```

```
(4191, 2)
```



Cluster Interpretation

```python
# Cluster Profiling (Most Important Step)
# Understand what each cluster represents.

cluster_profile = rfm.groupby('Cluster').mean()
print(cluster_profile)
```

```
            Recency   Frequency      Monetary
Cluster
0         42.755903    3.413542    837.452623
1          1.000000  148.000000  43518.310000
2         12.379310   18.444444   5327.811762
3        245.770554    1.519120    308.776789
```

Cluster	Recency	Frequency	Monetary	Customer Type
0	43 (moderate)	3.4 (low–moderate)	837 (moderate)	Regular Customers
1	1 (very recent)	148 (extremely high)	43,518 (very high)	VIP / Best Customers
2	12 (recent)	18 (good)	5,327 (high)	Loyal / High-Value Customers
3	246 (very high → long time ago)	1.5 (very low)	308 (low)	Lost / Inactive Customers

Business Interpretation

- **Cluster 1 — “VIP / Best Customers”**

These are our most valuable customers. They buy often, spend a lot, and have been active recently. To keep them engaged, we can offer loyalty programs and exclusive deals tailored to them.

- **Cluster 2 — “Loyal / High-Value Customers”**

These are strong, consistent customers, though not yet in the top tier. We can encourage them to become VIPs by using targeted upselling and cross-selling strategies.

- **Cluster 0 — “Regular Customers”**

These are average customers who purchase occasionally and are not highly engaged. We can target them with promotions and engagement campaigns to increase their activity.

- **Cluster 3 — “Lost / Inactive Customers”**

These customers have either churned or are at high risk of doing so, making them the biggest risk group. We can attempt to re-engage them through win-back campaigns, targeted discounts, or reactivation emails. If resources are limited, they may need to be deprioritized.

Monitoring and Updating

Customer behavior evolves over time, so clusters should be updated regularly to remain accurate. A monthly update is typically a good practice to capture these changes.

2. Demand Pattern Analysis

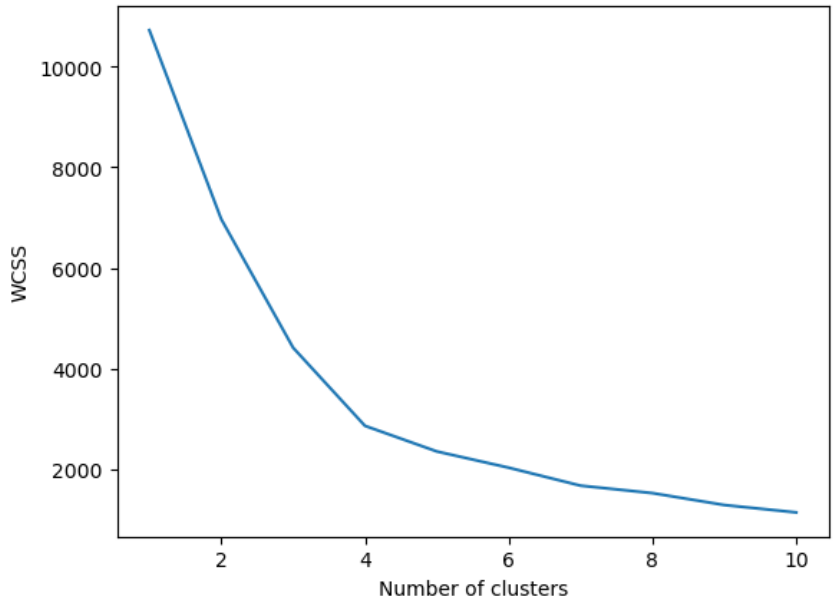
In this analysis, we cluster products (items) based on their demand patterns to gain a deeper understanding of how different items perform in terms of sales. To measure demand, we apply an adapted version of the RFM (Recency, Frequency, Monetary) framework at the product level rather than the customer level.

- **Recency** refers to how recently a product was last purchased. This metric helps identify whether an item is currently in demand or if it has not been sold for a long period, which may indicate declining relevance or seasonal behavior.
- **Frequency** represents how often a product is purchased, or the total number of units sold over a given period. This provides insight into the consistency and popularity of an item, helping distinguish between regularly purchased products and those with sporadic demand.
- **Monetary** reflects the total revenue generated by each product, highlighting its overall contribution to the business. This allows us to identify high-value items that significantly impact total sales.

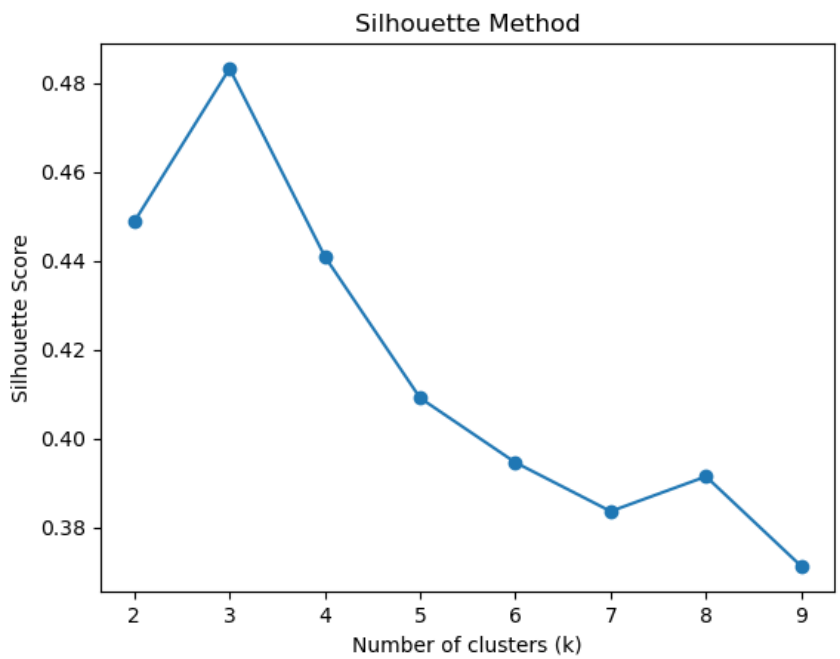
By combining these three metrics, we can group products into meaningful segments such as fast-moving items, consistently purchased products, and low-demand or slow-moving inventory. These segments provide valuable insights that support more informed decision-making in areas such as inventory management, demand forecasting, pricing strategies, and overall product portfolio optimization.

Determine the optimal number of clusters.

Elbow Method



Silhouette Score



The optimal number of clusters = 3

Cluster Visualization

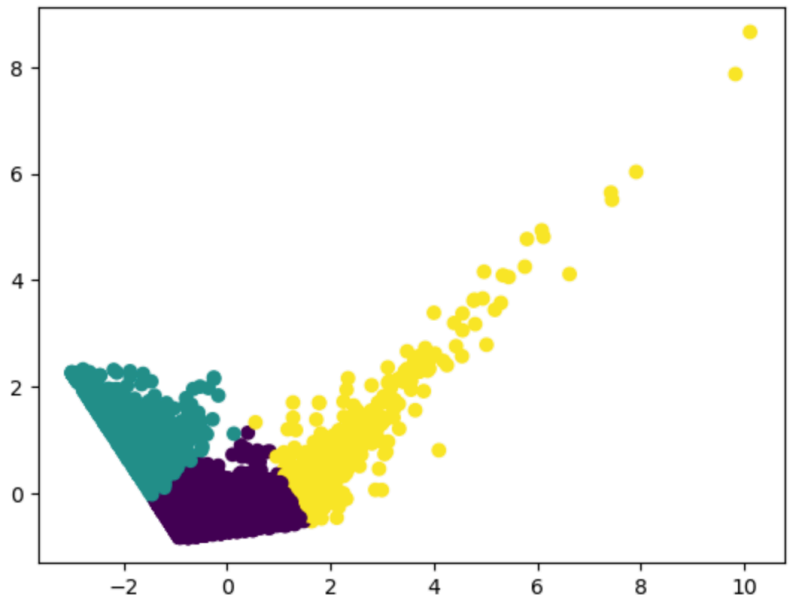
```
# Cluster Visualization
# Using PCA (dimensionality reduction)

from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

pca_product = PCA(n_components=2)
rfm_product_pca = pca_product.fit_transform(rfm_product_scaled)
print(rfm_product_pca.shape)

plt.scatter(rfm_product_pca[:,0], rfm_product_pca[:,1], c=rfm_product["Cluster"], cmap='viridis')
plt.show()
```

```
(3575, 2)
```



Cluster Interpretation

```python
# Cluster Profiling (Most Important Step)
# Understand what each cluster represents.

cluster_profile = rfm_product.groupby('Cluster').mean()
print(cluster_profile)
```

```
            Recency  Frequency     Monetary
Cluster
0         15.176970   6.686061   667.327525
1        231.436335   3.310559   138.631444
2          3.607456  13.164474  5610.881382
```

Cluster	Recency	Frequency	Monetary	Demand Pattern
0	15.2 (moderately recent)	6.7 (moderate repeat purchases)	667.3 (moderate spend)	Mid-tier demand items
1	231.4 (not purchased in a long time)	3.3 (low purchase rate)	138.6 (low revenue)	Slow-moving items
2	3.6 (very recent purchases)	13.2 (very frequent purchases)	5610.9 (very high spend)	Fast-moving items

Business Interpretation

- **Cluster 2 — “High-demand / Fast-moving items”**

Cluster 2 represents the top-performing, fast-moving items that are purchased very recently and frequently, generating the highest revenue. These are core products or bestsellers. We should prioritize keeping them in stock, use them for promotions and upselling, and forecast their demand carefully since they drive overall business performance.

Product Insights

Word frequency analysis of product names/descriptions, reveals patterns in which types of items are popular.
High-frequency words: BAG, SET, BOX, BOTTLE, LUNCH, CUTLERY, HOLDER

Cluster 2 is heavily focused on **functional everyday items**, especially:

- Storage (boxes, bags)
- Kitchen & dining (cutlery, lunch items, bottles)
- Giftable sets

These are **practical, repeat-purchase products**, which aligns well with high-frequency buying behavior.

Customers tend to prefer buying items in sets or packs, as they feel they get better value for money. There is also a clear liking for vintage and decorative styles, showing that people care about how products look, not just how they work. Colors like red, pink, blue, and white play an important role, as visually appealing products attract more attention and sales. Words like heart, love, and Christmas suggest that many items are bought as gifts, especially during special occasions and holidays. Overall, many products are used for home decoration and organization, meaning customers are often shopping to improve the look and feel of their homes.

- **Cluster 0 — “Steady / Mid-tier demand items”**

Cluster 0 represents consistent but not top-tier items that are purchased regularly, though not very frequently, and generate stable mid-level revenue. These are likely supporting or secondary products. We should maintain balanced inventory levels, explore cross-selling with high-demand items, and consider promoting them to grow their performance into top-tier products.

Product Insights

Word frequency analysis of product names/descriptions, reveals patterns in which types of items are popular.
High-frequency words: BOX, BAG, MUG, GLASS, PAPER, WRAP, COVER

Cluster 0 is heavily focused on decorative and gift-oriented items, especially:

- Functional items (bags, mugs, boxes)
- Gift-related packaging (wrap, paper)

Customers tend to buy many products as gifts for special occasions, like holidays, driven by emotional appeal, making them perfect for seasonal promotions and gift bundles. They also like stylish and decorative items, especially with vintage or floral designs, showing your brand’s strength in design-focused home décor. Home décor and ambient items, such as candles, wall hangings, and garden pieces, are popular as people decorate and enhance their living spaces. Color matters too, with bright, soft, and neutral tones like pink, blue, red, white, green, black, and silver influencing purchases. Finally, customers enjoy buying sets or bundles of multiple items, seeing them as better value, which helps increase basket size and overall sales.

- **Cluster 1 — “Low-demand / Slow-moving items”**

Cluster 1 represents slow-moving or declining items that haven’t been purchased recently and contribute little to overall sales. These products may be obsolete, seasonal, or unpopular. We should reduce inventory, use discounts or clearance strategies to move stock, and assess whether to discontinue or replace them with newer items.

Product Insights

Word frequency analysis of product names/descriptions, reveals patterns in which types of items are popular.
High-frequency words: HEART, ROSE, BRACELET, EARRINGS, CHARM, CHRISTMAS, EASTER

Cluster 1 is heavily focused on products designed for gifting or special occasions, appealing to emotional value. especially:

- Home decor items (ambient lighting, mirror, cushion)
- Accessories (jewelry items like bracelets, necklaces, and earrings)

This cluster represents products that are primarily gift-oriented, decorative, and lifestyle-focused, catering to customers who are looking for items that are both practical and visually appealing. Many of these products are designed with holidays and special occasions in mind, such as Christmas, Easter, or birthdays, making them popular choices for gifting. Customers are drawn to well-designed and aesthetically pleasing items, including vintage patterns, floral motifs, and playful designs, which show that style and visual appeal play a major role in purchasing decisions.

Additionally, products in this cluster are often sold in sets, bundles, or multi-piece packs, which adds perceived value and encourages higher spending per purchase. Bright, neutral, and metallic colors like pink, blue, white, red, gold, and silver are frequently chosen, highlighting the importance of color variety in attracting buyers. Functional everyday items such as bags, mugs, and glassware coexist alongside home

décor pieces like candles, cushions, mirrors, and decorative holders, providing a mix of practical use and lifestyle enhancement.

Overall, the cluster appeals to customers who are seeking gifting options, visually attractive home décor, and versatile lifestyle products, with demand driven by design, color, bundling, and emotional value. This makes it a high-potential segment for promotions, seasonal marketing campaigns, and bundling strategies aimed at maximizing sales and customer engagement.